Build, evidence and feasibility report

IIT Mandi Track B experimental workspace · Nepal flood case · 15 October 2026

Executive readout

The local flood workspace completes a bounded area/date workflow from Sentinel observations through candidate masks, historical infrastructure intersections, access uncertainty and evidence exports. Today's browser check confirmed saved-run selection, optical/SAR view choices, always-visible question controls, processing record and downloads. The UI exposes timestamped processing steps; no screen recording has been produced.

The frozen two-AOI comparison with UNOSAT Product 4259 is a warning, not a success claim: water-map agreement is weak (IoU 0.0134; recall 1.4%) against a preliminary, non-field-validated mixed event extent. Predictions were not retuned after scoring.

This supports an honest engineering demonstration only. Flood accuracy, debris classification and real settlement isolation are not established.

Measured evidence at a glance

Measure	Observed result	Scope
Regression suite	70 passed; 0 failed; 2.62 s; 31 warnings	Fresh run; deprecation/georeferencing warnings
Verified area/date export	Completed; 15 files hash-checked	event-fe47db331390
UNOSAT water-map agreement	IoU 0.0134; F1 0.0264; recall 1.4%	57,446 valid pixels over two AOIs; not ground truth
EMSR landslide proxy diagnostic	IoU 0.291; F1 0.450; recall 29.2%	47,127 pixels; class mismatch, not debris validation
Real road access	Unknown in automatic AOIs	Synthetic graph tests pass; real isolation unresolved

The two real-run count sets describe different footprints and must not be combined. The automatic Workspace example is a rectangular bounding box; nepal-syapru-004 clips to the irregular official AOI. Candidate values are not confirmed impacts.

1 · What was built

The flood lab runs at localhost:8770 with its own app view, results, evidence and local configuration. It reuses the established SatQuery visual foundation while keeping the flood workflow limited to area/date analysis. A local Qwen3-0.6B process classifies a question topic; fixed English/Nepali templates render saved map numbers and source pointers. The model does not generate measurements.

Workspace workflow

Stage	What happens	Evidence retained
1. Select event	Enter a past UTC event date and W/S/E/N bounds. Bounded STAC discovery looks for full-coverage Sentinel-1 dual-polarization acquisitions on matching platform/orbit, about 12 days apart.	Selected scene metadata; query plan
2. Process imagery	CDSE Sentinel Hub requests terrain-corrected GRD on a common 10 m projected grid. Scene identity, bands, masks, units, dates, orbit and grids are checked. Sentinel-2 L2A candidates are checked within ±14 days; three nearby dates per side are inspected.	Provider request, response hashes, observation metadata, optical coverage
3. Calculate candidates	A fixed SAR VV threshold/change rule marks new-water candidates. An optical vegetation-to-bare-ground plus SAR-change proxy marks debris candidates. RGB previews use B04/B03/B02.	GeoTIFF prediction, RGB previews, coverage and method descriptions
4. Assess infrastructure	Historical OSM features are intersected with candidates. Historical road ways retain original node IDs for a directed connectivity graph and access comparison.	OSM snapshot request, feature tables, graph audit, unknown statuses
5. Freeze and explain	Calculations and exports are frozen with hashes. Copilot routes the question and inserts values from saved result JSON.	15 downloadable files in the verified full job; citations to source fields

Current interface

The question, answer language and topic choice stay visible above area/date controls. Saved runs show evidence-backed answers, selectable optical RGB and SAR views, and impact layers. During processing, the dark-and-gold card shows elapsed time and timestamped backend events. The Processing Record lists inputs, masks, calculations, infrastructure snapshot and frozen outputs. Settlement access, evidence/downloads and limitations are foldable. Browser verification checks interface behavior, not map accuracy.

Local launch and saved evidence

PowerShell: `Clone or open the repository folder; run .\start-demo.ps1`
Verified 15-file area/date export: event-fe47db331390. Current test record: docs/test-results.xml; raw inputs and frozen outputs under data/ and artifacts/ are local-only and excluded from this public source branch. Keep credentials in ignored .env.local.

2 · Data and references used

Primary inputs and their roles

Source	Use in this work	Limits / attribution
Copernicus Data Space Ecosystem (CDSE), Sentinel Hub	Sentinel-1 GRD and Sentinel-2 L2A discovery/processing for the Nepal event and tested area/date runs.	Contains modified Copernicus Sentinel data 2026. Free, full and open use is subject to the Sentinel Data Legal Notice; CDSE processing quotas apply.
Copernicus DEM GLO-30 via AWS open data	Elevation source for provider terrain processing; public Copernicus DEM tile used for the corridor.	DSM, not bare-earth terrain; data has its own licence/terms and geographic tile coverage caveats.
HeiGIT ohsome historical extraction + OpenStreetMap historical API	July 27 corridor snapshot for official AOIs; event-date-minus-one-day snapshot for automatic area/date run. Historical way versions provide road node topology.	OSM data is ODbL and requires attribution/share-alike treatment for derivative databases. HeiGIT v2 API keys have a free tier; quota applies.
Copernicus EMS activation EMSR927, supplied GRA GeoPackages	AOI01 Syapru Besi and AOI02 Timure reference geometry for checking only. AOI01 landslide polygons compared after prediction freeze.	Evaluation-only. No reference geometry was used for prediction or threshold tuning. Reference class and image dates differ from proxy.
Qwen/Qwen3-0.6B	Local English/Nepali topic routing for the evidence copilot.	Pinned revision c1899de289a04d12100db370d81485cdf75e47ca; Apache-2.0. Not a flood segmentation model.

Official specification and data-source URLs

Evaluation-only external map source

UNOSAT Product 4259 / FL20260826NPL was evaluated only after prediction freeze. The retrieved archive is recorded in evaluation_only/references/unosat-fl20260826npl-source.json; its SHA-256 is 1c09bd5cff4e5dee870cd8413af0eb036e0dcd75d254cf1a337aa49096aaec2e. Product is preliminary and not field-validated.

IIT Mandi hackathon organizer / Track B rules
https://multimodal-ai-hackathon-2026-7.devpost.com/

CDSE home and API overview
https://dataspace.copernicus.eu/

CDSE Sentinel Hub service/quota context
https://dataspace.copernicus.eu/ecosystem/services/sentinel-hub

CDSE quotas
https://documentation.dataspace.copernicus.eu/Quotas.html

HeiGIT ohsome API v2
https://docs.ohsome.org/ohsome-api/v2/migration_guide.html

OpenStreetMap attribution and ODbL
https://www.openstreetmap.org/copyright

Copernicus DEM GLO-30 AWS registry
https://registry.opendata.aws/copernicus-dem/

Official EMSR927 activation
https://mapping.emergency.copernicus.eu/activations/EMSR927/

UNOSAT Product 4259
https://unosat.org/products/4259/

Qwen3-0.6B model card / licence
https://huggingface.co/Qwen/Qwen3-0.6B

3 · Observed results and validation

Official AOI run · nepal-syapru-004

Output	Saved value	Interpretation
New-water candidate pixels / area	361 / 36,100 m²	Fixed uncalibrated SAR rule; not confirmed flood extent
Water in both observations	87 pixels	Not evidence of permanent water
Mapped / intersected buildings	355 / 1	Historical OSM features; intersection is exposure only
Road length / bridge features	87.40 m / 2	Road/bridge candidate intersections; not confirmed closure
Debris candidates	2,123 pixels / 212,300 m²	Untrained vegetation loss to bare ground plus SAR proxy
Connectivity	Not assessed for official AOI; settlements unknown	The graph result retains AOI boundary, coverage and snap uncertainty

Automatic area/date run · event-fe47db331390

Measure	Saved result
Event date / sampled box	26 August 2026 · Syapru Besi rectangle
Raster dimensions	254 × 437 at 10 m; 110,998 pixels
Paired-valid SAR	99,858 pixels (89.96%)
Water candidates	956 pixels / 95,600 m²
Infrastructure	522 mapped; 2 intersected; 185.85 m road/bridge linework
Debris proxy	1,041 pixels / 104,100 m²; not validated
Access	Unknown: no hubs snapped; not zero isolated settlements
Optical search	Three nearby dates per side inside 14 days; greatest valid overlap chosen without reference labels
Frozen artifacts	15 downloads matched recorded SHA-256 hashes

Second official AOI · Timure

Measure	Saved result
Run / imagery	event-02163045d752 · Sentinel-1D 16/28 August, same ascending orbit 85
Valid pixels	82,653 / 99,540
Water candidates	143,900 m²; 4 intersected buildings
Road / bridge intersections	195.20 m; 1 bridge feature
Access	Unknown: no hubs snapped for the sampled rectangle

Independent-reference results and limits

Predictions were frozen and hashes checked before scoring; no thresholds were retuned. Across two disjoint AOIs, masked to official AOIs, UNOSAT analysis extent and valid prediction pixels, water comparison: 57,446 pixels; TP 318, FP 531, FN 22,906, TN 33,691; IoU 0.0134, F1 0.0264, precision 0.3746, recall 1.4%. This is poor external map agreement, not validated flood accuracy. Product 4259 is preliminary and not field-validated; its extent is not a water-only label, and sensors, dates, resolution and mapping methods differ.

The debris proxy vs EMSR927 class “6-Mass Movement / Landslide” is only a class-mismatch diagnostic: 47,127 pixels; TP 4,425, FP 64, FN 10,740, TN 31,898; IoU 0.291, F1 0.450, precision 0.986, recall 29.2%. This is not debris validation.

The 12-question development router check scored 9/12 (75%). Errors included casualty routing and Nepali access/confidence routing. Native Nepali review remains outstanding.

4 · Test results and runtime

Latest Python regression suite: 70 tests, 0 failures, 0 errors, 0 skipped; 2.62 seconds in the captured JUnit report. Pytest also reported 31 warnings in its console summary; warning categories were not saved in the JUnit XML.

Test area	Count	Examples checked
Core input contracts	14	Pre/event/post, orbit, units, grids, masks, role and hashes
Impact and graph logic	11	Bridge cuts, alternate route, overpass, one-way roads, boundary/unknown coverage
Acquisition + automatic jobs	13	Area bounds, pair selection, validation, duplicate jobs, restart/failure handling
Pipeline and evaluation	12	End-to-end exports, repeatability, tiled pixelwise equality, tamper/grid/reference guards
Copilot, server, debris, historical OSM, registration	20	Evidence pointers, unsupported claims, model answers, OSM historical nodes/access, debris proxy and reference registry

Runtime measurements

Metric	Measured	Meaning
Local calculate/export	1.36 s mean, n=13, range 0.70–2.26 s	Saved results; excludes catalogue, remote acquisition and processing.
Complete Workspace requests	50.6 s mean, n=6, range 35.5–81.8 s	Queued to saved result; 4 Syapru rectangle runs and 1 Timure request. Small sample.
Test suite	2.62 s	70 automated checks; tests acquire no imagery.
Transferred run inputs/artifacts	25.8 MB	One job folder; excludes model cache and other project data.

Run-time conclusion

Local raster/export stages averaged 1.36 seconds across 13 saved results. Complete area/date jobs averaged 50.6 seconds across 6 runs (35.5–81.8 seconds); four repeat the Syapru rectangle and one covers Timure. This small sample is not a general service average. Provider response, quotas, network, cloud cover and optical search affect wall time. The app bounds requests to one million 10 m pixels and 2500 pixels per side.

5 · Expense, feasibility and impact

Expense and operating assumptions

Cost item	Evidence-based status	What is not included
Sentinel / CDSE processing	Official CDSE general-user Sentinel Hub access is free within the published per-user quotas. We did not retrieve an account invoice or quota-use receipt.	Commercial/high-volume processing, excess capacity, quota changes
HeiGIT / ohsome	API key registration is free; quota and fair-use limits apply. No bill receipt was available.	Higher quota or commercial special arrangements
OSM historical data	OpenStreetMap data under ODbL; attribution and database obligations apply. No per-run data purchase recorded.	Compliance and distribution work
Copernicus DEM	AWS registry describes public GLO-30 access as free to the general public under its terms.	Network/ISP charges, other cloud egress or compute
Qwen3-0.6B	Apache-2.0 model; weights downloaded once (~1.2 GB per local model record).	Local GPU/CPU purchase, electricity, storage
Local run overall	No invoice or metered electricity/network/engineering-cost records were present. External service charges incurred are therefore not independently known. Expected direct data/API cash cost: $0 while staying in the public/free quotas and terms.	Engineering hours, local hardware depreciation, power, internet, paid quota, taxes

Feasibility

Demonstrated for two bounded Nepal AOIs and a separate Syapru rectangle: authenticated Sentinel imagery, optical/SAR pairing, historical OSM, impact overlays, graph construction and exports. Five completed area/date jobs averaged 50.6 seconds: four Syapru rectangle runs and one Timure request. The UNOSAT comparison is poor and not validated accuracy. The baseline is feasible to run locally, but not as a validated operational flood product. Scaling and broad-area runtime remain unmeasured.

Benefits and possible impact

Demonstrated workflow benefit: scene selection, raster preparation, historical OSM matching and evidence exports run in one traceable flow; saved calculations and hashes let analysts inspect how numbers were produced. This reduces handoffs in the prototype, but time savings have not been measured.

Potential impact: a validated future system could help analysts prioritize field checks and review infrastructure exposure. No agency deployment, user study, response-time reduction, lives saved, verified damage or actual evacuation/road-closure decision was measured.

Material risks and closure items

Water agreement with preliminary UNOSAT extent is poor (recall 1.37%); do not present candidate area as mapped flood extent. The reference is not field-validated and includes mixed flow types. Debris remains an untrained vegetation-loss/SAR proxy; the landslide comparison cannot validate debris. Synthetic graph tests cover bridge cuts, alternate routes, overpasses and missing hubs, but real settlement isolation remains unknown where hubs, coverage or boundaries are unresolved. Nepali output needs native human review; no screen recording has been produced. Organizer reuse clearance and submission approval remain unconfirmed.

Readiness decision

The local workspace is ready for a candid engineering demonstration with the low UNOSAT agreement, unvalidated proxies and unknown access stated up front. It is not ready to claim validated flood/debris accuracy, reliable settlement isolation or response outcomes, and it is not submission-ready until reference, human-review and recording gaps are closed.

Appendix A - 90-second demo script

Before recording, open the final report PDF in a second window at page 5. Start at Workspace with event-fe47db331390 selected and the page at the top. Click cues are silent; read the quoted narration verbatim. Target duration is 1:30 maximum. The spoken script is about 142 words.

Time	Click while speaking	Say exactly
0:00-0:08	Workspace; confirm Saved analysis is event-fe47db331390.	This is SatQuery's separate flood workspace. I'm opening the saved Syapru run for 26 August.
0:08-0:21	View > Before optical RGB; click image to zoom, use wheel, press Esc, then select After optical RGB.	These are the available optical dates. The water calculation compares Sentinel-1 scenes from 16 and 28 August on orbit 85.
0:21-0:35	View > Impact layers; point to the candidate and exposure metrics.	This run marks 95,600 square metres of water candidates, two intersected buildings, 185.85 metres of roads, and two bridge features.
0:35-0:44	Point to the prefilled question and click Ask.	These indicate possible exposure, not confirmed damage. Answers use this run's saved calculations.
0:44-0:56	Scroll to Processing record and point to its seven steps; expand Settlement access.	The processing record lists seven steps. Access remains unknown because no hub was snapped; unknown is not proof of safety.
0:56-1:08	Click Evidence; point at the regression table and source/download links.	CDSE imagery and HeiGIT plus OpenStreetMap road history feed one traceable workflow. Seventy software tests passed.
1:08-1:20	Press Alt+Tab to the report PDF, open on page 5; point at the UNOSAT results.	Five full area/date runs averaged 51.6 seconds. UNOSAT agreement was poor: IoU 0.013, recall 1.37%, against a preliminary mixed extent.
1:20-1:30	Keep report page 5 visible.	Debris and real access remain unvalidated. This may help analysts prioritize review; costs and time savings are unmeasured.

Accuracy and safety cue: the UNOSAT result is poor map agreement against a preliminary mixed extent, not flood accuracy or field truth. Access is unknown when graph hubs or coverage are missing; it is not zero impact. The 51.6-second figure is a five-run local sample (35.5-81.8 seconds), not a service-wide average.